

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0778

Betzwood 138 kV
May 21, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0778. The Interconnection Customer proposes a 93.9 MW solar photovoltaic facility interconnecting at the Betzwood 138 kV substation in the Pennsylvania area. The request was received on April 19, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 344 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,449,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Fruitland - E Bench 138 KV	Starkey Road - Pico 138 KV	101.4	196	5.0
Kce Ny 1 - N Mobile 138 KV L	Owatonna - Moser Generating Sta Substation 345 KV	101.3	240	11.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fruitland - E Bench 138 KV	Reconductor line	\$544,000	16% (\$85,000)
Kce Ny 1 - N Mobile 138 KV L	Reconductor line	\$2,016,000	68% (\$1,364,000)
	Total allocated to project		\$1,449,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.